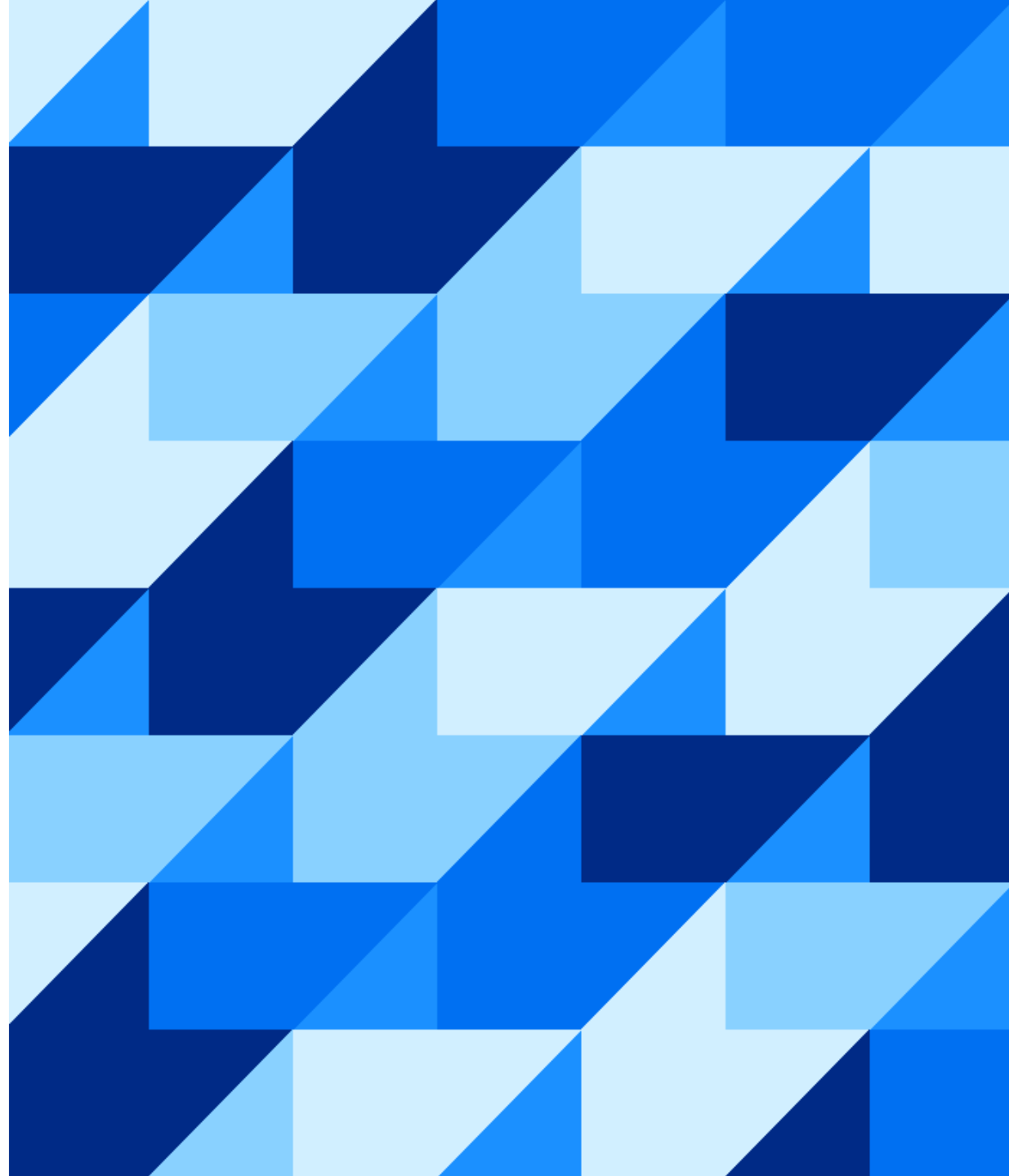


Tabular AI & SAP-RPT

Turning powerful tabular models to production-ready predictive capabilities

Partner Enablement

11 August 2026



Large language models
AI for unstructured data



New category of models

Tabular foundation models

AI for structured data

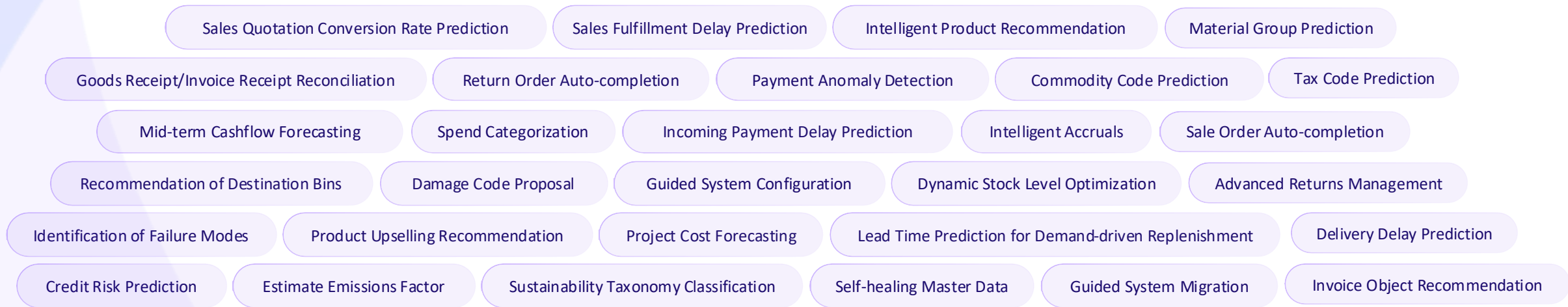


Large language models

AI for unstructured data



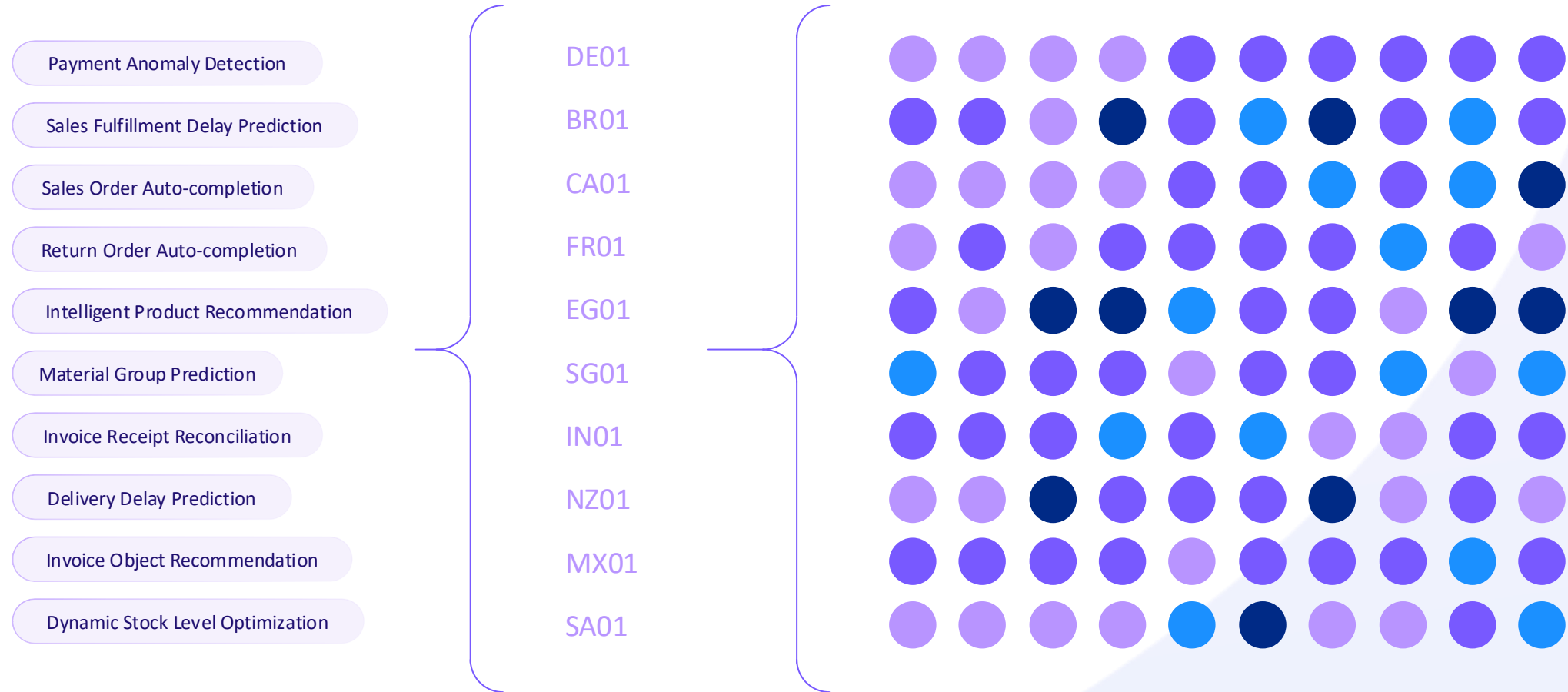
Predictions



10 Predictive tasks

10 Company codes

100 Models



1

Model



SAP-RPT

Relational Pretrained Transformer



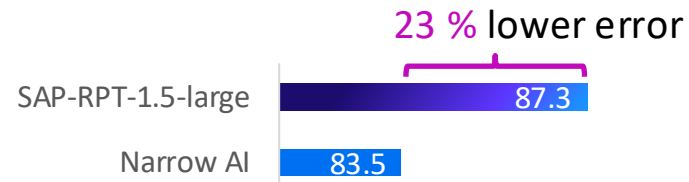
SAP-RPT - 1.5



SAP-RPT in a nutshell



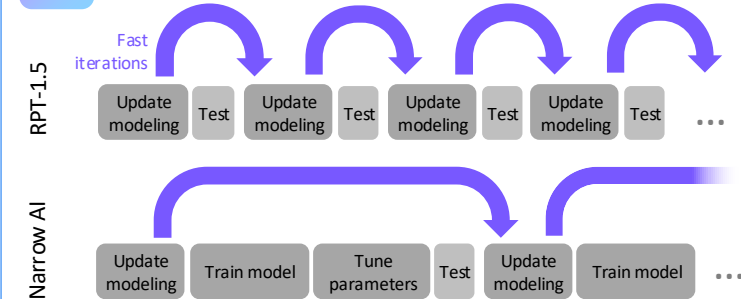
More accurate predictions



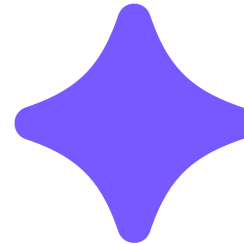
Average accuracy (↑) across 10+ SAP scenarios, 200+ customer datasets



Faster development cycles



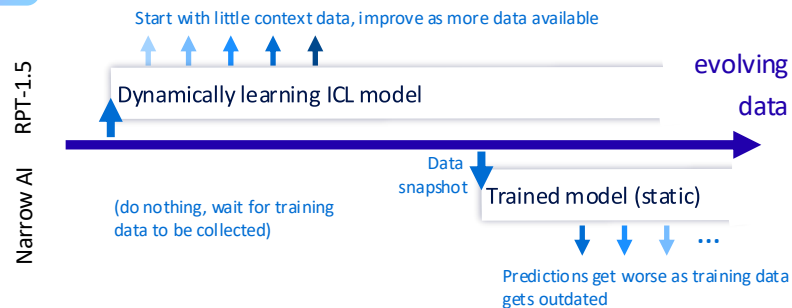
Unlocking
new use cases



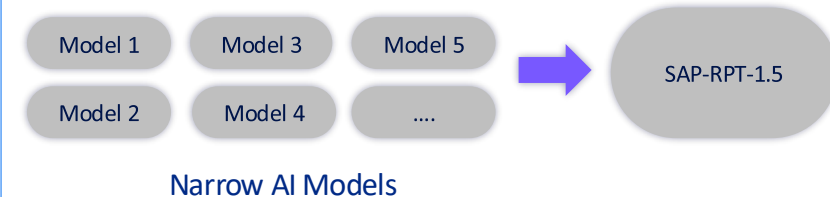
Democratizing &
Scaling Tabular AI



Speed-to-value & continuous learning



Kill model sprawl



In-Context Learning for Tabular Data

1

Provide examples and questions together

Input

The context

Invoice number	Line item	Supplier	Price	General ledger account
0001	1	Stark Industries	100	100
0001	2	Stark Industries	200	101
0002	1	Gringotts Bank	4000	100
...	...	...	...	...

The query /

Invoice number	Line item	Supplier	Price	General ledger account
0003	1	Soylent Corp	100	???
0003	2	Soylent Corp	200	???

Engine

SAP-RPT

2

Enable the model to understand the task from the context during a single, real-time forward pass

Output

Invoice number	Line item	Supplier	Price	General ledger account
0003	1	Soylent Corp	500	100
0003	2	Soylent Corp	1000	101

3

Get instant, task-specific predictions without any model retraining



SAP-RPT-1.5

► Now generally available on gen AI hub

- **Explainability**
 - Classical column-wise explanations (“feature importances”)
 - Novel context-row-based explanations (“explain by example”)
- **Improved performance** for large-scale data:
 - ~4.3pp – 5.9pp accuracy gains for high-cardinality use cases
 - ~7.5pp – 11.9pp F1 Macro score gains for large SAP datasets
- Context management & unlimited context via Tabular Orchestration (*GA planned for August 2026*)

```
"predictions": [  
  {  
    "COSTCENTER": [  
      {  
        "prediction": "Office Furniture",  
        "confidence": 0.96,  
        "confidence_interval": null  
      }  
    ],  
    "ID": "35"  
  }  
],  
"explanations": {  
  "top_column_scores": [  
    {"DESCRIPTION": 0.523, "AMOUNT": 0.234},  
    {"AMOUNT": 0.612, "DESCRIPTION": 0.188}  
  ],  
  "top_relevant_context_rows": [  
    [3, 4, 2],  
    [4, 3, 2]  
  ]  
},  
],
```

Detailed benchmarking results are published in our [model card](#)

More information can be found on our [Help Portal](#) page



SAP-RPT-1.5 Variants

SAP-RPT-1.5

Optimized for **speed**

Available via RPT playground

SAP-RPT-1.5-Large

Optimized for **accuracy**

No changes to pricing and model specifications.

RPT-1 and RPT-1.5 will be priced identically.

See [SAP Note](#) on model availability and cost. Use the [AI Core Calculator](#) for cost estimation.



SAP-RPT External Playground



[SAP-RPT Playground](#)



[ConTextTab Paper](#)

SAP-RPT Playground

DocumentationFeedback

Legal Disclaimer: This is a demonstration platform. By using this service, you agree to comply with applicable data protection regulations.

Welcome to the SAP-RPT Playground

Choose from pre-built data sets or upload your own CSV data to get instant predictions and insights powered by SAP's advanced machine learning models. Get started with our sample scenarios or explore the full capabilities of [SAP-RPT](#).

Show me demo scenarios

Joule



Welcome !

QUICK START

What is SAP-RPT?

How do I upload my data?

Find me a scenario

Ask or create anything

Joule uses AI, verify results.

Sample Scenarios

**Equipment Sourcing**

Procurement

Toy dataset to get you started. Estimate carbon footprint from equipment order details.

Try Now

**Service Ticket Prioritization**

Customer Service

Predicts the priority level of customer service tickets to accelerate support.

Try Now

**Payment Delay Prediction**

Finance

Prediction of the number of days payments are late to forecast cash flow, reduce risk.

Try Now

INTERNAL | 13

Tabular AI in SAP

RPT models are powerful model engines but use cases and business processes need more than just a component

They need a system that removes the complexity and handles the **heavy lifting** of turning a model into a reliable, governed business process

We envision Tabular AI to be the **vehicle** that supports the use of our core model engines

Key Objective 1

Unified interface for all (existing and future) predictive models

Key Objective 2

Extensive tooling to support easier use of predictive models

Key Objective 3

E2E build-test-deploy workflow to build and run any predictive use cases



Tabular Orchestration

- Available on gen AI hub as a new orchestration workflow for tabular foundation models (TFMs)
 - Default and production-ready integration pathway for RPT models and new TFMs
- Main mode of consumption: **API**

Key Features

- Centralized Context Management – connect your HANA table and the workflow can manage the data retrieval
- Automated Context Selection – select from available context selection strategies

▶ GA workflow with [targeted context selection](#) targeted for **Aug 2026**



Why use Tabular Orchestration?

Automated Context Management

Workflow handles the ingestion of context data and automates the selection of relevant context rows

Better predictions from using our **latest targeted context retrieval approach**, which gives an uplift over baseline context selection methods for RPT models

Compatibility

Tabular Orchestration is the **harmonizing layer** across existing and new tabular foundation models

Switch easily between TFMs via a common interface



Getting Started with Tabular Orchestration

Design-Time Steps

(Done once per use case)

- 1 Set up a data destination
- 2 Register your tabular artifact
- 3 Create your AI use case scenario

- 4 Create a virtual deployment of Tabular Orchestration

(Done once per service key, i.e., multiple scenarios can use the same virtual deployment)

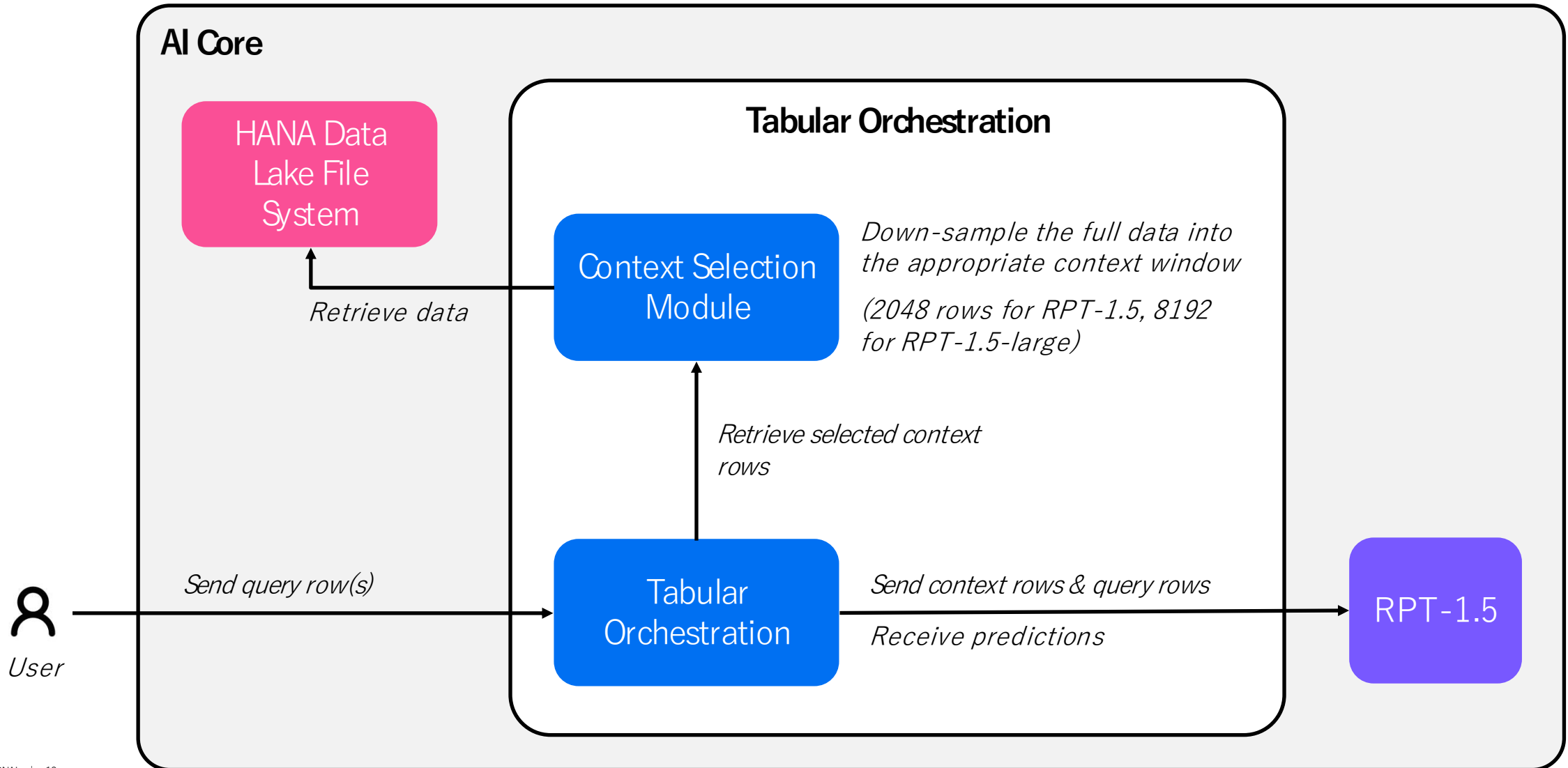
Run-Time Steps

(Call whenever you require predictions)

- 1 Call the Tabular Orchestration workflow with your defined AI use case scenario



How It Works



Let's get hands-on!

Hands-On: Introduction

In the exercise, we will use **Predicting Employee Flight Risk in HR** as an example use case.

Through the exercise, you will get to:

- Run the steps of setting up the Tabular Orchestration workflow for the example use case
 - Set up a data destination
 - Register your tabular artifact
 - Create your AI use case scenario
 - Create a virtual deployment for the workflow
- Run the workflow to get predictions from SAP-RPT-1.5 for the example use case



Hands-On: Getting Started

Head to the [WorkZone page](#) and download the zip file to get started.

The zip file contains:

- Raw Datasets
 - `Person_Employment_Info.csv`
 - `Succession_Export.csv`
 - `User_Directory.csv`
- Python notebook for the hands-on exercise (`Hands_On_Exercise.ipynb`)
- Credentials folder with HDLFS client certificate and keys as well as AI Core service key
 - *Note that credentials will only be available for 24 hours*
- Bruno collection for all API requests required
 - *Optional for the Hands-on. We provided a Bruno API collection in case you would like to run any individual API calls.*



Hands-On: Getting Started

Extract the zip file.

Open the extracted folder in your preferred code editor.

Setup a local environment using the provided `requirements.txt`. If you are new, you can follow the instructions [here](#).

Start the local environment.

Open the Python notebook.

Run the code blocks sequentially.



Hands-On: Context Selection Strategies in Tabular Orchestration

This can be toggled via the `contextSelectionConfig.strategy` parameter, which is an optional parameter

Supported values include:

- `auto`: workflow will use heuristic by default, fallback to random if there is an error (e.g., timeout)
- `heuristic`: fetch context rows semantically closest to each query row, fails if there is an error (e.g., timeout)
- `random`: sample context rows at random (fast, good baseline)
- If no value was provided, the workflow will use random



Hands-On: Important Notes

- For this exercise
 - All participants will be sharing a common HDLFS instance on our Canary landscape
- Make sure that all names used when creating AI Core components are in lowercase alphanumeric with hyphens only
- RPT-1.5 is already generally available for you to try out.
 - Refer to the [help portal guide](#) for more details.
- Tabular Orchestration will only be generally available from early September 2026.
 - The exercise was created using the sap-internal plan. You should create an AI Core under the **extended** plan when you are setting it up on your own.
 - You will be able to follow the same steps in your own setup once it is available.
 - The workflow currently supports up to 32 query rows per call.
 - The workflow currently does not yet return the explanations from RPT-1.5. This will be supported at the time of GA.



Hands-On: Extension

As an extension, you can adjust the provided code for your own use case.

Duplicate the provided notebook. Use your own dataset(s) and edit the data preparation logic as required. You can go on to upload your context data to the common HDLFS instance.

Make sure to not upload sensitive data. Ensure that there are no PII's and data have been properly masked / anonymized.

To get a feel of the full benefit of the targeted context selection strategy, we recommend you using datasets that are much larger than the max context windows for the model variants

- 2048 context rows for RPT-1.5
- 8192 context rows for RPT-1.5-large



Q&As

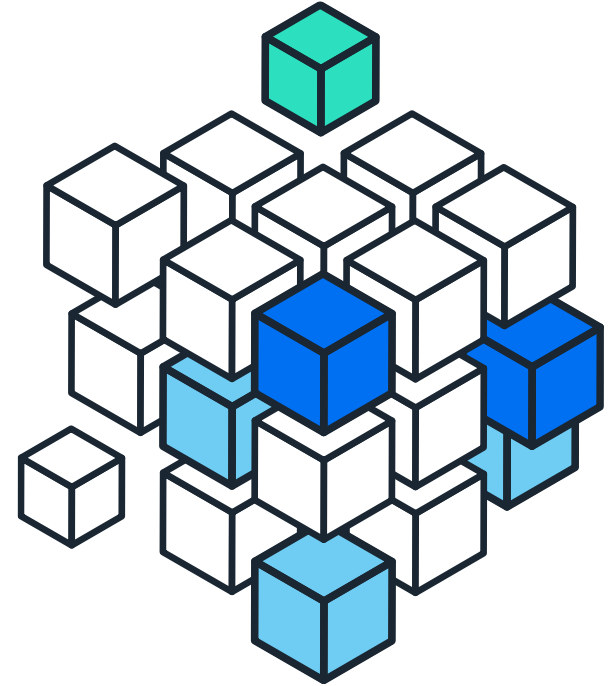
Thank you.

Contact information:

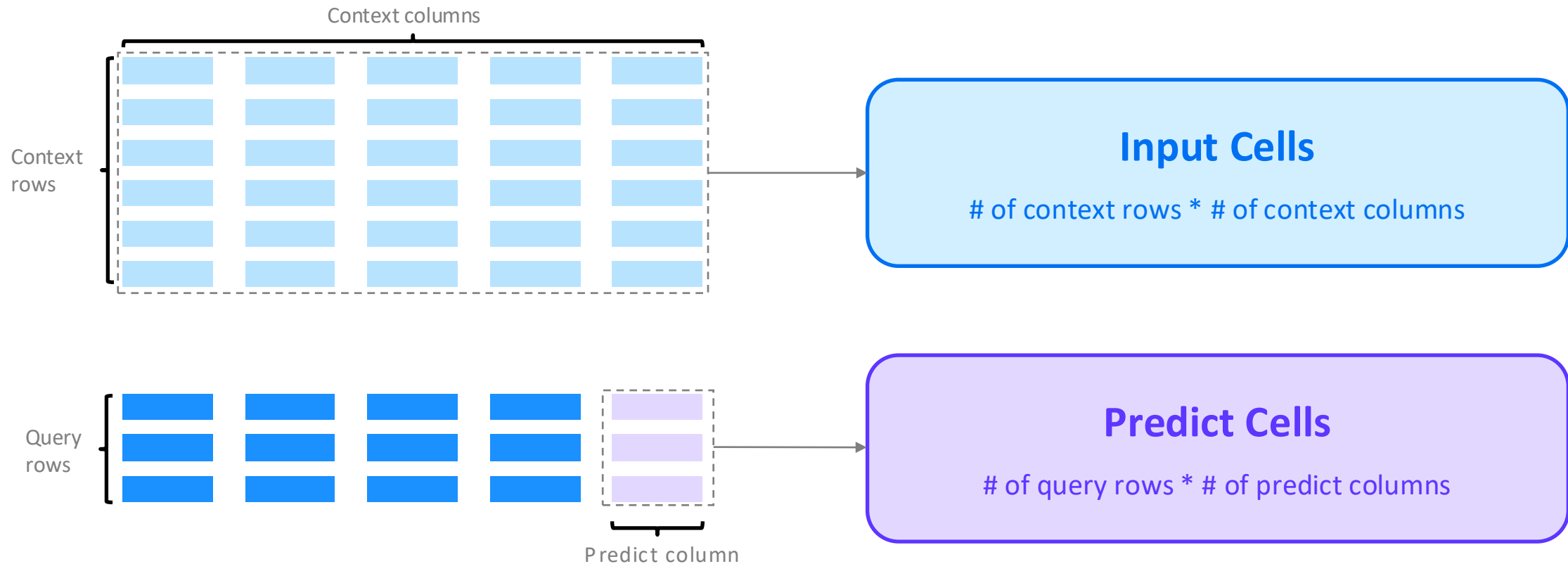
Sherene Tan

Product Manager for Tabular AI / SAP-RPT

sherene.tan@sap.com



How RPT Pricing Works



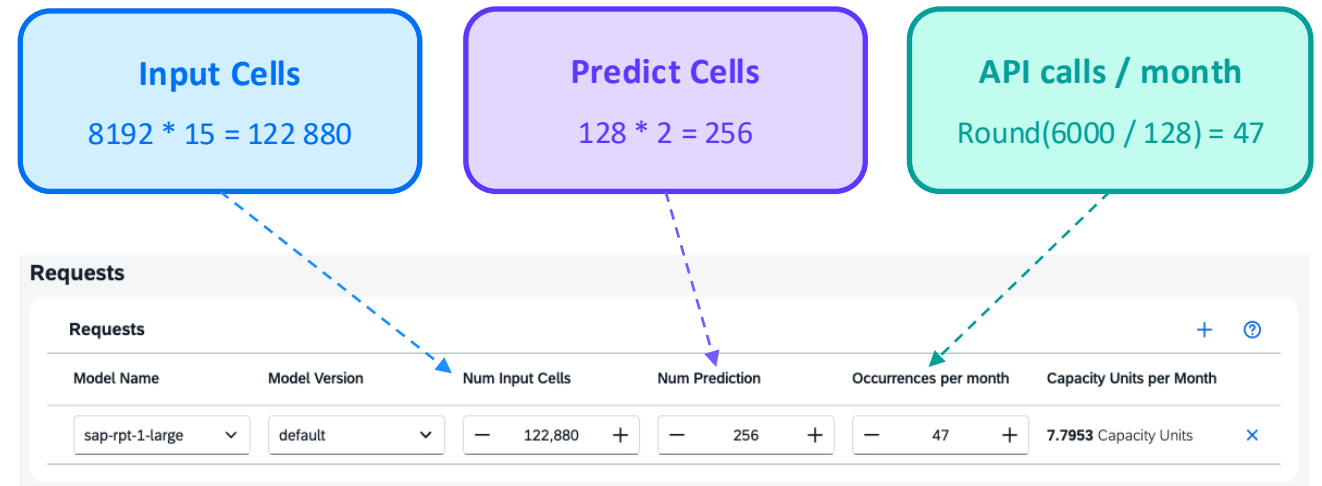
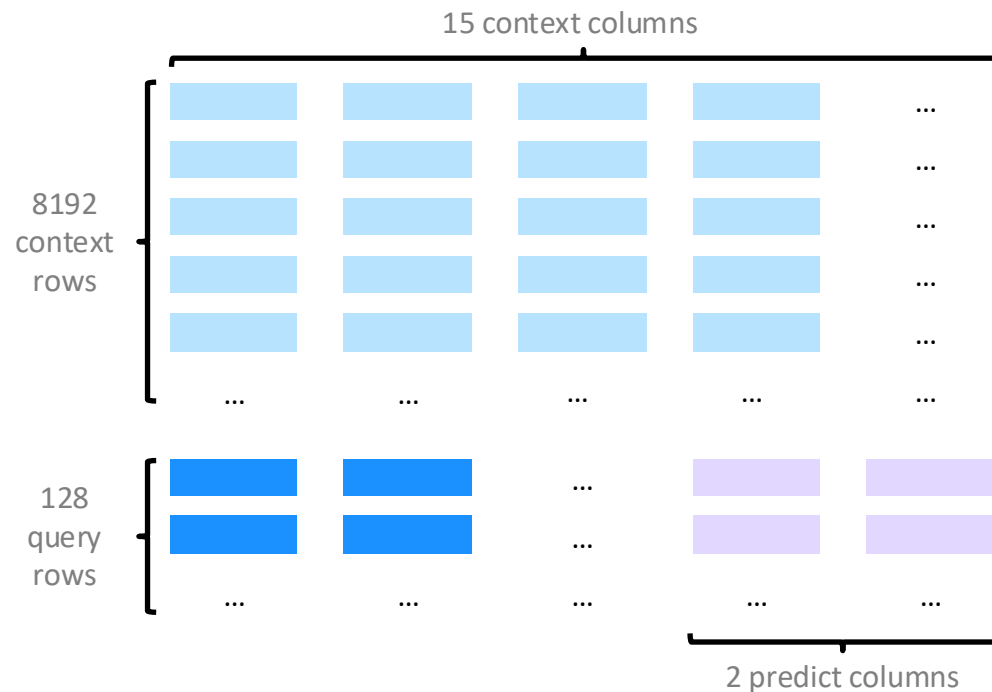
Cost per API = Cost of input cells + Cost of predict cells



How RPT Pricing Works

Example Use Case: Sales Order Completion

- 8192 historical records with 15 features
- 128 query rows per API, each with 2 predict cells
- Assume 6000 sales order per month
- Using SAP-RPT-1.5-Large



Setting Up AI Core on BTP

Entitlements

Ensure that your Global Account admin has granted the following entitlements to your subaccount:

- ai-core
- ai-launchpad

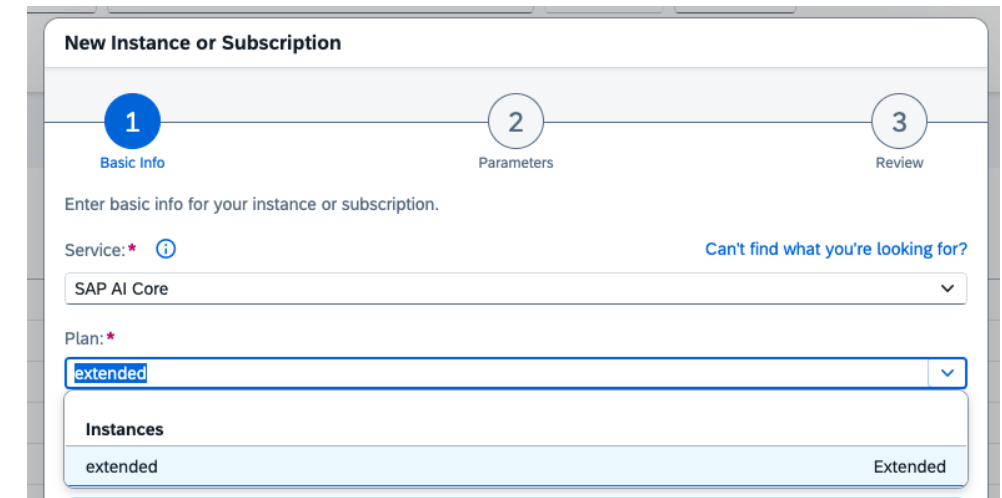
Quota is dependent on your specific needs



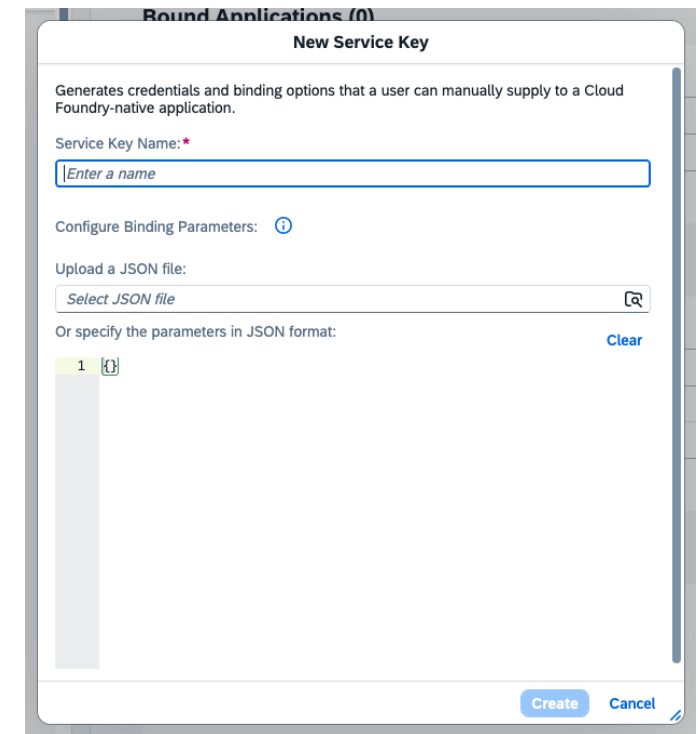
Setting up AI Core

Make sure to create an AI Core instance under the **extended** plan

Once the instance is created, under the **Service Key** section, click **Create** to create a service key. Give it any name you'd like and click **Create**.



The screenshot shows the 'New Instance or Subscription' form with three steps: 1. Basic Info, 2. Parameters, and 3. Review. The 'Basic Info' step is active. It prompts the user to 'Enter basic info for your instance or subscription.' The 'Service' dropdown is set to 'SAP AI Core'. The 'Plan' dropdown is set to 'extended'. Below the dropdowns, there is a table with the heading 'Instances' containing one entry: 'extended' with the status 'Extended'.



The screenshot shows the 'New Service Key' form. It includes a description: 'Generates credentials and binding options that a user can manually supply to a Cloud Foundry-native application.' The 'Service Key Name' field is empty with a placeholder 'Enter a name'. Below this is the 'Configure Binding Parameters' section, which includes an 'Upload a JSON file' button with a file icon and a 'Clear' button. At the bottom, there is a 'Create' button and a 'Cancel' button.



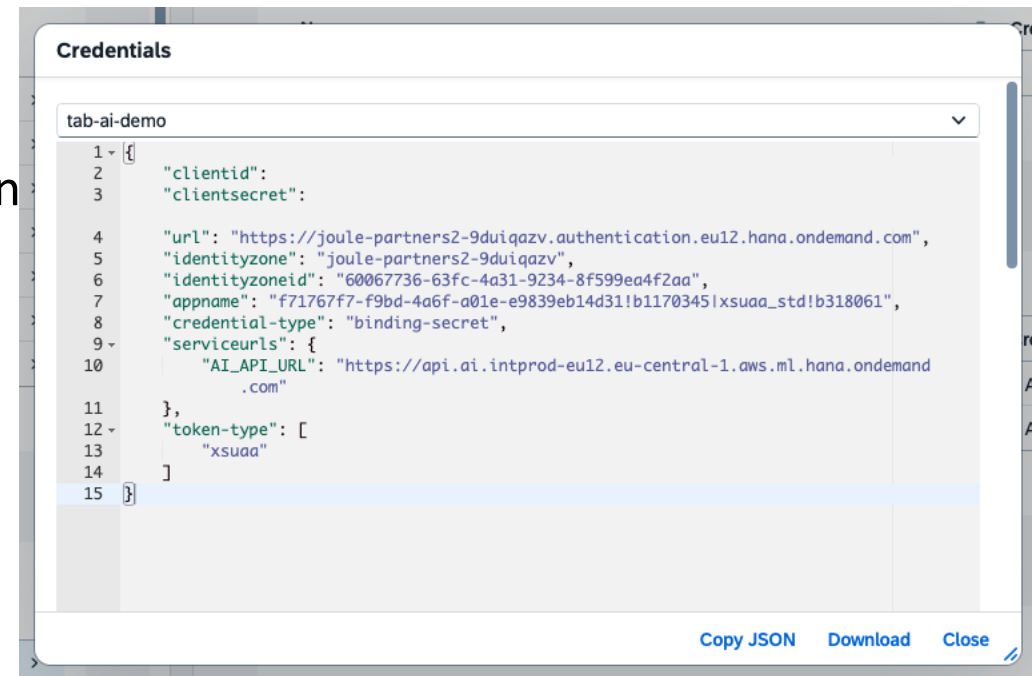
Obtaining AI Core Service Key

Once the service key is created, you will see that it is JSON with several key-value pairs

Note down the following:

- `url`, `clientid` and `clientsecret` – required to obtain the bearer token
- `AI_API_URL` – the base URL for any calls to API AI Core

The bearer token is needed for every API call to AI Core



The screenshot shows a 'Credentials' window with a dropdown menu set to 'tab-ai-demo'. The main area displays a JSON object with the following structure:

```
1 {  
2   "clientid":  
3   "clientsecret":  
4  
5   "url": "https://joule-partners2-9duiqazv.authentication.eu12.hana.ondemand.com",  
6   "identityzone": "joule-partners2-9duiqazv",  
7   "identityzoneid": "60067736-63fc-4a31-9234-8f599ea4f2aa",  
8   "appname": "f71767f7-f9bd-4a6f-a01e-e9839eb14d31!b1170345!xsuaa_std!b318061",  
9   "credential-type": "binding-secret",  
10  "serviceurls": {  
11    "AI_API_URL": "https://api.ai.intprod-eu12.eu-central-1.aws.ml.hana.ondemand  
12    .com"  
13  },  
14  "token-type": [  
15    "xsuaa"  
16  ]  
17 }
```

At the bottom right of the window, there are three buttons: 'Copy JSON', 'Download', and 'Close'.



Setting Up HDLFS

Entitlements

Ensure that your Global Account admin has granted the following entitlements to your subaccount

- hana-cloud > relational-data-lake
- hana-cloud-tools > tools

Quota is dependent on your specific needs

The screenshot displays the SAP HANA Cloud entitlements configuration interface. At the top, there is a search bar with the text 'hana' and a 'Show auto-entitled services' checkbox which is checked. Below the search bar, the 'Data and Analytics' section is expanded, showing two sub-sections: 'SAP HANA Cloud | hana-cloud' and 'SAP HANA Cloud | hana-cloud-tools'. The 'hana-cloud' section contains a table with three rows: 'hana' (unchecked), 'hana' (unchecked), and 'relational-data-lake' (checked). The 'relational-data-lake' row shows a quota of 2. The 'hana-cloud-tools' section contains a table with one row: 'tools' (checked), showing a quota of 3. Both sections have a 'Cloud Foundry' label in the top right corner.

Plan	Available in Regions	Quota
☐ hana	Europe (Frankfurt) - Canary - AWS; Europe (Frankfurt) - AWS	1
☐ hana	Europe (Frankfurt)	1
☒ relational-data-lake	Europe (Frankfurt) - Canary - AWS; Europe (Frankfurt) - AWS	2

Plan	Available in Regions	Quota
☒ tools	US East (Virginia) GCP Canary; US Central (IA) G...	3



Setting up HDLFS

Create an SAP HANA Cloud application and a HANA instance under relational-data-lake plan

Once created, access the dashboard and you should already see 1 instance running

SAP

HANA Cloud Central

Instances

Migrations

Schemas & HDI Containers

Tenants

Alerts

SQL Console

Database Objects

Data Lake Files

Import and Export

Data Products Monitor

Performance Monitor

Performance Analyzer

Try quick actions like create instance or show SQL console

Q

ST

All Instances

Create Instance

Search:

State:

Alerts:

Type:

Version:

Adapt Filters

Instances (1) Subaccount Roles (28)

Group Instances

↓↑

State	Name	Type	Version	Alerts	Runtime Environment	Memory	Storage	Compute	Nodes	Replicas	Encryption Key	Created On	Actions
hana-dl													
Running	hana-dl	Data Lake Data Lake Relational Engine Component Enabled	2026.2.7363-00 (QRC 2/2026)		Cloud Foundry			2 vCPU Coordinator 2 vCPU Workers (1)			SAP-Managed	Jul 23, 2026, 06:03:45 PM	⋮ >

INTERNAL | 36

Setting up HDLFS

Access the instance and click **Manage Configurations**

Under the **Connections** section, select **Allow all IP addresses**

Connections

Connections

If you select to only allow SAP BTP, then you will not be able to connect using tools such as dbisql.

Allowed connections: *

☐ Allow only IP addresses from Cloud Foundry in this BTP region

☒ Allow all IP addresses

☐ Allow specific IP addresses and IP ranges (in addition to Cloud Foundry in BTP)

AWS PrivateLink Endpoint IDs:

Establish connections between services in your Virtual Private Cloud (VPC) and your data lake instance.

☐



HDLFS Endpoint

You can find the HDLFS endpoint which is required for registered a data destination via the Data Lake Files tab

SAP

HANA Cloud Central

Try quick actions like create instance or show SQL console

ST

Instances

Migrations

Schemas & HDI Containers

Tenants

Alerts

SQL Console

Database Objects

Data Lake Files

Import and Export

Data Products Monitor

Performance Monitor

Performance Analyzer

Data Lake Files

Last refreshed at 19:18:56

Data Lake Files

Search: State:

Adapt Filters

File Containers (1)

State	Name	Single Sign-On Enabled	Endpoint	Actions
Available	hana-dl		27863fd0-157b-40c8-8869-4fd0c9fbd4.files.hdl.prod-eu12.hanacloud.ondemand.com	

INTERNAL | 38

Important Note 1

When you first create a data destination to your HDLFS instance, you will get 201 created with the register and the **mTLS subject patterns** generated for this destination

Add these subject patterns to your HDLFS instance. Access the instance and click **Manage Configurations**.

Under the **Authorizations** section and add 2 new authorizations as follows. Each subject pattern will be 1 authorization.

Authorizations

Create authorizations to define what specific clients are allowed to do once their trust is validated. Authorizations match a client's subject to a regular expression pattern and assign roles to that client. Authorizations with lower rank values are evaluated first. Trust bindings are the associations between trusted client certificates and specific authorizations.

Authorizations (3)

Add

Rank	Roles	Trust Bindings	Pattern	
— 1 +	user x		CN=HANA_63e96ac1-a225-4f61-88c3-0a9527fa0ca7_eu12,L=hcm:hc:hana:prod:dev:tc,OU=fcf4333e-6822-45f4...	Generate Pattern x
— 1 +	user x		CN=ab9fddf2ce4a0e60d84ddd076eee367a7c0e30f98ca5929feb5484e8098cd38a,L=dac4af2a7ec165985c8c535...	Generate Pattern x



Important Note 2

Before you can upload any new files to your HDLFS instance, you will need to generate a certificate to make a secure connection

You can follow along [Step 3](#) and [Step 4](#) in the developer guide to set this up

